General Mathematics — Exercise Sheet 23

Antiderivatives and integration rules

Variant: Statements

Exercise 1 (Constants and a missing point). Let $D = \mathbb{R} \setminus \{0\}$ and $f : D \to \mathbb{R}$ be given by $f(x) = |x|/x$. Find all primitives of f on D. Select the primitive satisfying $F(-1) = F(1) = 2$. Determine whether this selected function has a continuous extension to zero, and whether such an extension can be differentiable there. Use Lecture 7's one-sided derivative test and Lecture 23's interval theorem.

Exercise 2 (Substitution and the original domain). Find complete primitive families for the following integrands.

1. The trigonometric integrand $\tan^2 x/\cos^2 x$ on $D_1 = \{x \in \mathbb{R} : \cos x \neq 0\}$. State which integration constants are independent on its domain intervals.
2. The logarithmic integrand

$$\frac{\ln x}{x(1 + (\ln x)^4)} \qquad (x > 0).$$

   State whether $x = 1$ requires a separate constant or is a regular point of the same primitive family.

Exercise 3 (Parts with a logarithmic factor). Find all primitives of $x \ln x$ on $(0, \infty)$ by integration by parts. Then select the primitive F satisfying $F(e) = 0$. State the factors u and v' you use and verify the selected answer by differentiation.

Exercise 4 (Two logarithmic candidates). On $D = (0, 1) \cup (1, \infty)$, consider the integrand $1/(x \ln x)$ and candidates

$$F(x) = \ln|\ln x|, \qquad G(x) = \ln|x \ln x|.$$

Differentiate both candidates to determine which is a primitive. Give every primitive on D and explain whether an added constant can repair the failed candidate. Compare the missing point $x = 1$ here with its role in the preceding logarithmic substitution problem.

Exercise 5 (Sine-power reduction). Define $F_0(x) = x$ and $F_1(x) = -\cos x$ on $\mathbb{R}$. For integers $n \geq 2$, derive and prove the recurrence

$$F_n(x) = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} F_{n-2}(x)$$

constructing a primitive of $\sin^n x$. Use the product rule underlying integration by parts, with $\cos^2 x = 1 - \sin^2 x$, to justify the recurrence without assuming an unproved primitive already exists. Compute F_4 and give every primitive of $\sin^4 x$ on $\mathbb{R}$. Here $\sin^0 x$ means the constant function 1.